

INTERESTS I am a first-year Ph.D. student in Computer Science at Purdue University, advised by Prof. Jason Wu. Combining my background in design with computational methods, I design AI interactions for designers, focusing on *agentic systems*, *collaborative sensemaking*, and *generative interfaces* for creative work across video, UI, graphics, and motion. Recently, I have been bringing machine learning into the interaction loop so that AI can adapt to designers’ workflows.

EDUCATION	Purdue University, West Lafayette, IN	Sep 2026 - Present
	- Ph.D. in Computer Science - Advisor: Prof. Jason Wu	
	Korea Advanced Institute of Science and Technology, Daejeon, Korea	2020 - Feb 2026
	- B.S. in Computer Science and Industrial Design - Summa Cum Laude (Ranked 3rd, GPA: 4.08/4.3)	
	New York University, New York, NY	Spring 2024
	KAIST-NYU Minor in Integrated Design & Media (GPA: 4.0/4.0)	

PUBLICATIONS

CONFERENCE PAPERS

[C1] **VIVID: Human-AI Collaborative Authoring of Vicarious Dialogues from Lecture Videos**
Seulgi Choi, **Hyewon Lee**, Yoonjoo Lee, Juho Kim
ACM CHI 2024

POSTERS AND DEMOS

[P4] **Hangulo: Demonstrating Workflow-Embedded AI Support for Korean Lettering Implementation**
Hyewon Lee, Tak Yeon Lee
ACM UIST Adjunct 2026

[P3] **CORAL: Continually Learning Generative Interfaces from Collective User Interactions**
Daniel Lee, DaEun Choi, **Hyewon Lee**, Haijun Xia
ACM UIST Adjunct 2026

[P2] **RADI: A Design Framework for Relational and Adaptive Disclosure Interfaces**
Hyewon Lee*, Ihchae Ryu*, Yumin Cho*, Hyunseung Lim, Hwajung Hong
ACM UIST Adjunct 2025

[P1] **CamARa: Exploring and Creating Camera Movements with Spatial Reference in Augmented Reality**
Hyewon Lee, Christopher Bannon, Andrea Bianchi
ACM CHI Extended Abstracts 2025

PREPRINTS AND ARCHIVES

- [A3] **GUIDE: Designer-in-the-loop Authoring of Conformant Generative User Interfaces**
 Hyewon Lee, Ziying Wang, Aiden Moy, Saran Nagubandi, Jason Wu
arXiv 2026
- [A2] **“When to Hand Off, When to Work Together”: Understanding Concurrent Human-Agent Interaction in Shared Co-Creative Workspaces**
 Kihoon Son, Hyewon Lee, DaEun Choi, Yoonsu Kim, Tae Soo Kim, Yoonjoo Lee, John Joon Young Chung, HyunJoon Jung, Juho Kim
arXiv 2026
- [A1] **An Augmented Reality Camera Layout System for Intuitive, Spatially Aware Shot Design**
 Hyewon Lee, Jee Youn Kim, Christopher Bannon, Andrea Bianchi
Preprint 2025

RESEARCH EXPERIENCE

KIXLAB, KAIST Daejeon, Korea Fall 2023; Jul 2025 – Aug 2026
Research intern

- Advisor: **Prof. Juho Kim**
- Researched human–AI co-authoring of pedagogical dialogues for instructor support [C1].
- Exploring synchronous co-design interactions with in-tool generative agents [A2].

Graduation Research, KAIST Daejeon, Korea Mar 2025 – Dec 2025
Undergraduate researcher

- Advisor: **Prof. Tak Yeon Lee**
- Investigating human–AI collaborative vector editing workflows for Korean lettering through designer-centered sensemaking artifacts [P4].

MakeLab, KAIST Daejeon, Korea Aug 2024 – Jun 2025
Research intern

- Advisor: **Prof. Andrea Bianchi**
- Developed an AR-based camera tool bridging 3D scenes and real environments to support expressive cinematography [P1] [A1].

WORK EXPERIENCE

SketchSoft Seoul, Korea Jul 2022 – Dec 2022
Software engineer intern

- Developed a 3D sketching tool Feather (Three.js) and implemented GLTF curve transformation reducing file size by 87%.
- Proposed and designed a movie export interaction allowing users to record and share 3D sketches, extending the tool’s expressive capabilities.

SPARCS, KAIST Daejeon, Korea Mar 2021 – Dec 2024
Software engineer & UX designer

- Designed and implemented *Biseo*, an internal web service supporting voting and decisions for KAIST Programmers Association, later adopted by two additional campus groups.
- Improved UX as a design lead by resolving key usability issues and introducing an atomic design system for scalable front-end development.

AWARDS & HONORS

Korean Government Scholarship Program for Study Overseas (KGSPSO),
 Government of the Republic of Korea 2026

- Awarded to support doctoral study abroad; \$40,000 scholarship.

	Ross Fellowship with Herbold Scholarship , Purdue University · Awarded to incoming Ph.D. students; \$10,000 scholarship.	2026
	Next-Generation Engineer Award , Institute for Promotion of Engineering and Science of Korea, IPESK · Recognized as an outstanding student for excellence in research.	2025
	Undergraduate Research Program , KAIST · Selected for Undergraduate Research Program (URP); \$5,000 research grant.	2024
	College of Engineering Dean's List , KAIST · Achieved a perfect GPA of 4.3/4.3 (2nd in department); \$800 scholarship.	2022
	Lim Misuk Scholarship , KAIST · Recognized for leadership and societal contribution; \$1,000 scholarship.	2022
	Presidential Science Scholarship , Ministry of Science and ICT, Republic of Korea · Awarded by the President of Korea to outstanding science and engineering students. · Covered full tuition (4 years) and \$20,000 scholarship.	2020
TEACHING & MENTORING	Research Mentor , AFE · Mentored high school students on research projects in computer science.	2025 - 2026
	Undergraduate Tutor , KAIST · Taught data structures and algorithms; nominated as Best Tutor (2023).	2022 - 2023
	Teaching Assistant , KAIST · Taught CS101: Introduction to Programming (Python) lab sessions.	2021
SERVICES & LEADERSHIP	Student Volunteer , ACM SIGCHI Korea Local Chapter	2025
	Taekwondo Club Organizer , KAIST	2024 - 2025
	International Student Buddy , KAIST	2024
	Department President , School of Computing, KAIST	2022
	Cohort Representative , School of Computing, KAIST	2021